A square hollow structural section (150×150×8) ~~is used as a~~ tension member and is welded to a gusset plate as shown in the figure. Compute the design strength of the connection. S355 steel is used for all the members. Do not consider strength of the welds. In order to facilitate construction of the hollow structural section on the gusset plate, the hollow structural section is cut by additional 2 mm more than the thickness of the gusset plate.

S355

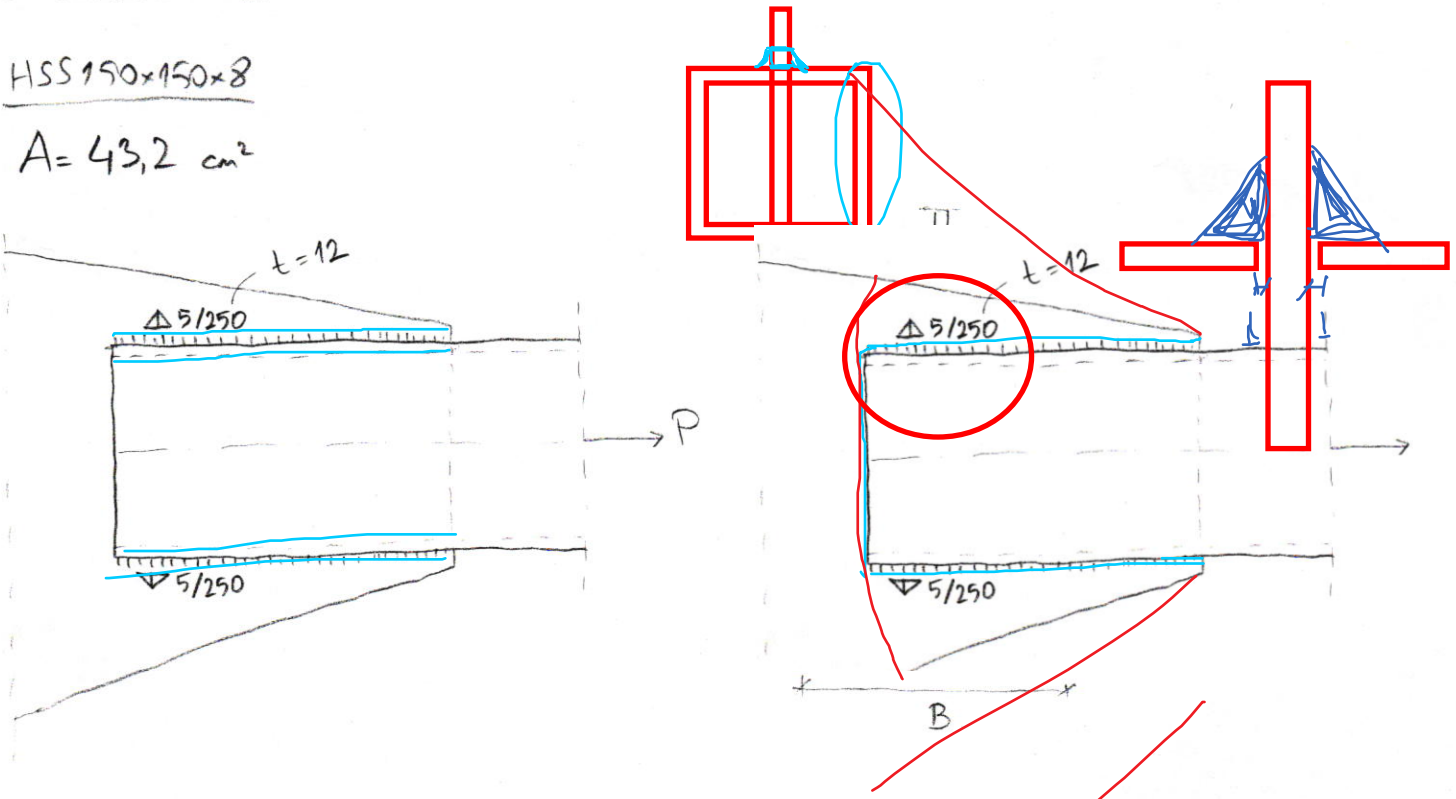
$F_y = 35,5$ kN/cm²

$F_u = 51$ kN/cm²

$E = 20000$ kN/cm²

HSS150×150×8

$A = 43,2$ cm²



→ Yielding: $T_n = F_y \cdot A_g = 35,5 \cdot 43,2 = 1533,6$ kN

$\phi_t T_n = 0,90 \cdot 1533,6 = 1380,24$ kN

→ Fracture: $T_n = F_u \cdot A_e$ $\quad A_e = U \cdot A_n$

$\ell \geq H \rightarrow U = 1 - \frac{\bar{x}}{\ell}$ $\quad \ell = 250$ $\quad H = 150$ $\quad \bar{x} = \frac{B^2 + 2BH}{4 \cdot (B+H)}$

$\bar{x} = \frac{150^2 + 2 \cdot 150 \cdot 150}{4 \cdot (150+150)} = 56,25$ mm $\quad U = 1 - \frac{56,25}{250} = 0,775$

$A_n = 43,2 - 2 \cdot (1,2+0,2) \cdot 0,8 = 40,96$ cm² $\quad A_e = 0,775 \cdot 40,96 = 31,74$ cm²

$T_n = 51 \cdot 31,74 = 1618,74$ kN $\quad \phi_t T_n = 0,75 \cdot 1618,74 = 1214,06$ kN

± Block Shear = (Case -1)

$R_n = 0{,}60 F_u A_{nv} + U_{bs} F_u A_{nt} \leq 0{,}60 F_y A_{gv} + U_{bs} F_u A_{nt}$

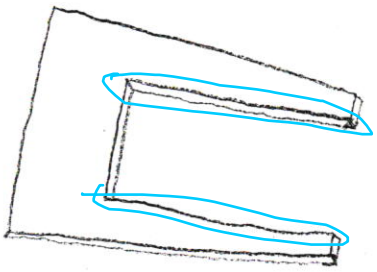
$A_{gv} = A_{nv} = 2.25.1{,}2 = 60\ cm^2$

$A_{nt} = 15.1{,}2 = 18\ cm^2$

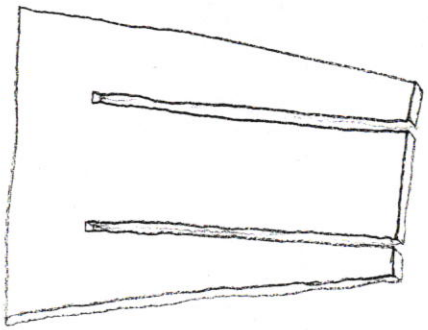
$R_n = 0{,}60.51.60 + 1.51.18 = 2754\ kN > 0{,}60.35{,}5.60 + 1.51.18 = 2196\ kN$

$R_n = 2196\ kN$

$\phi R_n = 0{,}75.2196 = 1647\ kN$



(Case -2)



$A_{gv} = A_{nv} = 4.25.1{,}2 = 120\ cm^2$

$A_{nt} \approx 0$

$R_n = 0{,}60.51.120 + 0 = 3672\ kN > 0{,}60.35{,}5.120 + 0 = 2556\ kN$

$R_n = 2556\ kN$

$\phi R_n = 0{,}75.2556 = 1917\ kN$

(Case -3)

